

# Elanchezhian Muthukumar

Software Developer

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## SKILLS

- **AI ML and Computer Vision:** Python, PyTorch, LLMs (GPT, Gemini AI Studio, Claude), Computer Vision (SAM, Detection, OCR), NLP (NER, Multilingual)
- **Workflow Automation:** N8N
- **Data and MLOps:** Pandas, NumPy, Feature Engineering, Model Deployment, Monitoring, CI/CD
- **Backend and APIs:** FastAPI, Django, REST API, Docker (Basic)
- **Cloud:** AWS, Google Cloud Platform
- **Frontend and DB:** Next.js, JavaScript, PostgreSQL, MongoDB, Supabase

## EDUCATION

- **PERIYAR UNIVERSITY**  
MASTER OF COMPUTER APPLICATIONS — University Rank Holder  
Aug 2022 – Jun 2024 | CGPA: 8.5
- **VYSYA COLLEGE**  
BACHELOR OF SCIENCE IN COMPUTER SCIENCE  
Jul 2019 – Jun 2022 | CGPA: 7.7

## CERTIFICATIONS

- Full Stack Web Developer – Udemy
- AI for India 2.0 – Skill India (Govt. of India)
- Programming Certifications – NPTEL
- Innovation & Entrepreneurship Bootcamp – MIC, AICT

## AWARDS AND HONORS

- **First Place** – Project Expo, Emerge 2K23, SRM Valliammai Engineering College (Nov 2023)
- **Consolation Award** – In-House Project Expo 2023, Periyar University (Sep 2023)

## PROFESSIONAL EXPERIENCE

### SOFTWARE DEVELOPER

MAHIMA TECHNOLOGY PRIVATE LIMITED | SALEM, TAMIL NADU, INDIA  
Jul 2024 – Present

- Engineered a **production ML pipeline** using **Google Gemini AI** for invoice data extraction across 15+ formats, achieving **96% accuracy** and reducing manual effort by **50%** while processing 800+ documents daily.
- Implemented a **Meta SAM-based document segmentation system** for layout-aware region detection, improving structured data extraction accuracy by **18%**.
- Architected and deployed a **file automation and validation platform** with alert-driven exception handling, reducing reconciliation time by **40%**.
- Constructed a **dual QC/QA PDF annotation tool** using Python and Django, enabling side-by-side comparison and **systematic discrepancy tracking**, improving early defect detection by **15%**.

### WEB DEVELOPMENT INTERN

RAKUMURA IT SOLUTIONS | CHENNAI, TAMIL NADU, INDIA  
Jan 2024 – Apr 2024 (During MCA)

- Led end-to-end development of responsive client websites, improving user engagement by **40%**.
- Collaborated on **10+ feature launches**, contributing to a **25% increase** in quarterly sales.

## PROJECTS

### SECURE AI DOCUMENT & BOOKMARK VAULT

TECHNOLOGIES: NEXT.JS, GEMINI AI, TESSERACT.JS OCR, CRYPTOJS, AES ENCRYPTION, GOOGLE DRIVE API | [\[LINK\]](#)

- Built a **personal zero-knowledge encrypted vault** for storing daily notes, bookmarks, and documents as a privacy-focused alternative to Google Keep and Google Drive.
- Implemented **OCR-based document indexing** with 96% accuracy, enabling fast search and retrieval across 800+ stored files.
- Integrated **Gemini AI-powered natural language commands** for intelligent search and organization, reducing manual information retrieval time by **2.5 hours/week**.

### VIRTUAL ASSISTANT FOR INDIVIDUALS WITH DISABILITIES

TECHNOLOGIES: PYTHON, FLASK, REST APIs, JAVASCRIPT, HTML5, CSS3 | [\[LINK\]](#)

- Created an **AI-powered desktop assistant** enabling hands-free system control, improving interaction efficiency by **50%**.
- Engineered a **modular Flask backend** to support speech and image-based automation workflows.

### EMPLOYEE MANAGEMENT SYSTEM

TECHNOLOGIES: MONGODB, EXPRESS.JS, REACT.JS, NODE.JS, JWT | [\[LINK\]](#)

- Implemented **RBAC for three user roles** (Admin, Manager, Employee) to secure sensitive HR data.
- Delivered a **JWT-secured REST API** with 20+ endpoints, achieving sub-150ms response times.
- Streamlined HR workflows through a self-service dashboard, reducing administrative workload by **40%**.